

# DICKSON OSEI TUTU

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## PROFILE

I am a passionate leader and computer science graduate with a background in software development and youth empowerment. Expedient in leading diverse teams, facilitating group dialogues, and driving initiatives that create positive change, I combine technical expertise in algorithms, data structures, and full-stack development with exceptional communication and leadership skills.

## EDUCATION

### BSc. Computer Science

Kwame Nkrumah University of Science and Technology (KNUST), Kumasi, Ghana

Expected September 2025

### W.A.S.S.C.E (General Science)

Prempeh College, Kumasi, Ghana

September 2018 – September 2021

## WORK EXPERIENCE

### Intern | GIT Plus Academy

3<sup>rd</sup> October 2024 - 1<sup>st</sup> December 2024

- Designed and developed a transaction history component using PHP.
- Gained hands-on experience in software development and problem-solving.
- Collaborated with the development team to implement the component.

## ACHIEVEMENTS/AWARDS

Outstanding Student Leader

24<sup>th</sup> August 2023

Computer Science Society, KNUST

## PROJECTS/RESEARCH EXPERIENCE

### Quiet Place: AI-Based Mental Health Web App (MERN Stack) - [Link](#)

- Developed a web application for booking appointments with counselors and interacting with an AI-powered bot.
- The bot recommends resources based on user interactions, providing personalized support.
- Utilized MongoDB, Express.js, React.js, and Node.js to build a seamless user experience.

### Retrieval-Augmented Generation (RAG) Mental Health Chatbot - [Link](#)

- Developed a Generative AI (GenAI) RAG system to create a context-aware mental health chatbot, providing accurate information grounded in PDF documents.
- Engineered data pipelines for document loading (PDFs) and efficient text chunking using LangChain's PyPDFLoader and RecursiveCharacterTextSplitter.
- Implemented semantic search capabilities by generating and managing Hugging Face embeddings (sentence-transformers/all-MiniLM-L6-v2) and storing them in a Pinecone vector database.
- Integrated Google's Gemini API (gemini-1.5-flash-latest) as the core Large Language Model (LLM) within a LangChain framework, leveraging its capabilities for grounded text generation.
- Developed a Flask web application to serve the RAG chatbot, including routing (@app.route) for the frontend and an API endpoint (/get) for handling user queries.

### Telegram Chatbot ( Python, Telegram Bot API) - [Link](#)

- Engineered a conversational Telegram bot capable of understanding and responding to user queries for mental wellness support.
- Designed and implemented core bot functionalities including command handling, group message filtering, and dynamic text-based responses.
- Ensured secure handling of API tokens by leveraging environment variables, enhancing deployment security and flexibility.

## Ghana Labour Information System - Software Testing | 2025

- Participated in a testing event, analyzing system functionality and reporting usability issues.
- Provided feedback to developers on UI/UX improvements and potential bugs.
- Gained hands-on experience in QA processes and stakeholder collaboration.

## PROFESSIONAL TRAINING/ CERTIFICATION

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|                                      |                                |                            |
|--------------------------------------|--------------------------------|----------------------------|
| Arduino Programming                  | 9 <sup>th</sup> May 2025       | TME Education              |
| Fiber Optics                         | 14 <sup>th</sup> February 2025 | Johvic Fibertech Solutions |
| Capacity Development Summit          | 10 <sup>th</sup> February 2024 | LEC Groups                 |
| Cybersecurity Certification          | 25 <sup>th</sup> July, 2025    | ISC2                       |
| Artificial Intelligence Fundamentals | 30 <sup>th</sup> July, 2025    | IBM SkillsBuild            |

## LEADERSHIP EXPERIENCE

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### Deputy Publicity Head (July 2025 - Present)

Prempeh College Old Students Association (Amanfoc), Ashanti Branch.

### Prayer Rep, Ushering Department (August 2024 - Present)

Pentecost Students and Associates (PENSA), KNUST

- Coordinate weekly prayer sessions, fostering spiritual growth and community cohesion.

### Chairperson, Science and Innovation Committee (December 2024 - Present)

Science Students' Association (SCISA), KNUST

- Lead organizer of SICE'25, the college's flagship science fair, overseeing a committee of 30+ members to showcase 18 innovative projects from students.
- Revitalized 5 dormant STEM clubs in the College of Science through targeted recruitment campaigns and mentorship programs.

### Vice Presidential Candidate (2024/25 Academic Year)

Students' Representative Council (SRC), KNUST

### President (April 2022 – February 2025)

Old Students' Association of Jesus Grace Academy

- Launched a leadership institute to build proactive leaders for the students and school.

### Secretary (December 2023 - January 2025)

Association of Past Students of Prempeh College and Yaa Asantewaa Girls' SHS (Amanadehyee), KNUST

- Increased member engagement by 40% by revamping the association's social media notification system.
- Coordinated high-profile events, including workshops, guest lectures, and conferences.

### Chairperson, Residential Committee (October 2023 – October 2024)

National Union of Ghana Students (NUGS), KNUST

- Engaged with the Zonal Police Department and Campus Security Department to advocate for improved safety measures for students both on and off campus.
- Spearheaded the distribution of first aid boxes to all traditional halls on campus, ensuring prompt medical assistance for residents.
- Advocated for improved residential facilities through discussions with university management.

### Representative, University Students' Association of Ghana, USAG (October 2022 – October 2023)

Kwame Nkrumah University of Science and Technology, KNUST

- Organized an educative campaign, USAG At Your Doorstep, to promote awareness and understanding of USAG and its existence.
- Led collaboration with Hive Ghana to host a seminar on decentralized web apps, managing logistics, promotion, and audience engagement for 300+ attendees.

## Assistant Senior Prefect (September 2020 – July 2021)

Prempeh College

- Identified, reviewed, and selected vendors to sell at specific times in accordance with class hours and prep times.
- Secured a first aid box for the Red Cross Society.
- Made the public address system available to student leaders.

## VOLUNTARY/EXTRA-CURRICULAR ACTIVITY

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- **NUGS-KNUST Community Outreach ( April 2024)** – Supported the team in transporting donations and other supplies to various destinations during the outreach program.
- **USAG Road Saftey Campaign ( December 2022)** - Educated drivers, passengers, and pedestrians on safe road practices.

## Technical Skills

- **Programming Languages:** JavaScript (ES6+), Python, HTML5, CSS3, PHP
  - **Frameworks & Libraries:** React, Node.js, Express.js, Next.js
  - **Databases:** MongoDB
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## Additional Strengths

- **Problem-Solving:** Proven ability to analyze complex technical challenges and develop effective, scalable solutions.
- **Full-Stack Development:** Experience building and maintaining web applications from front to back, leveraging the MERN stack (MongoDB, Express.js, React, Node.js).

## REFEREES

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